BIOCHEMISTRY CUME
January 24, 2003

A mouse is fed a diet including the amino acid glutamate and the fatty acid palmitate, but no carbohydrate. Small amounts of the essential amino acids and the necessary vitamins and minerals are also included in the diet.

1. (30) Can the mouse synthesize glucose on this diet using glutamate and palmitate as the sole source of carbon and energy? Explain your answer.

2. (40) If plamitate is removed from the above diet, can the mouse synthesize glucose using glutamtate as the sole source of carbon and energy? If yes, outline a metabolic pathway for this process.

3. ( ) If glutamate is removed from the above diet, can the mouse synthesize glucose using palmitate as the sole source of carbon and energy? If yes, outline a metabolic pathway for this process.

4. (30) The mouse is fed the original diet containing both unlabeled glutamate and palmitate, and then palmitate radiolabeled with C-14 at carbon 2 is added to the diet. Will glucose become labeled with C-14? If yes, give a complete metabolic pathway for the conversion of carbon 2 in plamitate to the carbon framework of glucose. Which carbon atom(s) of glucose will first become labeled?

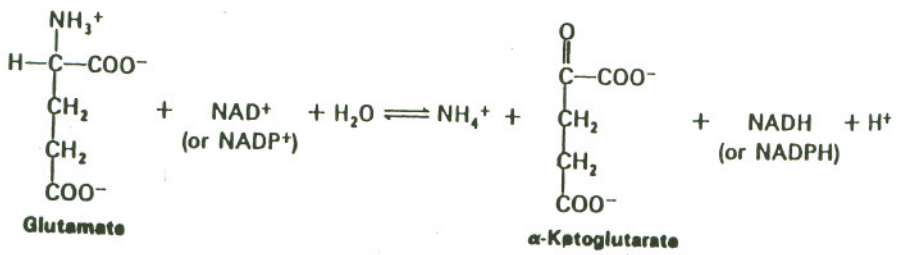
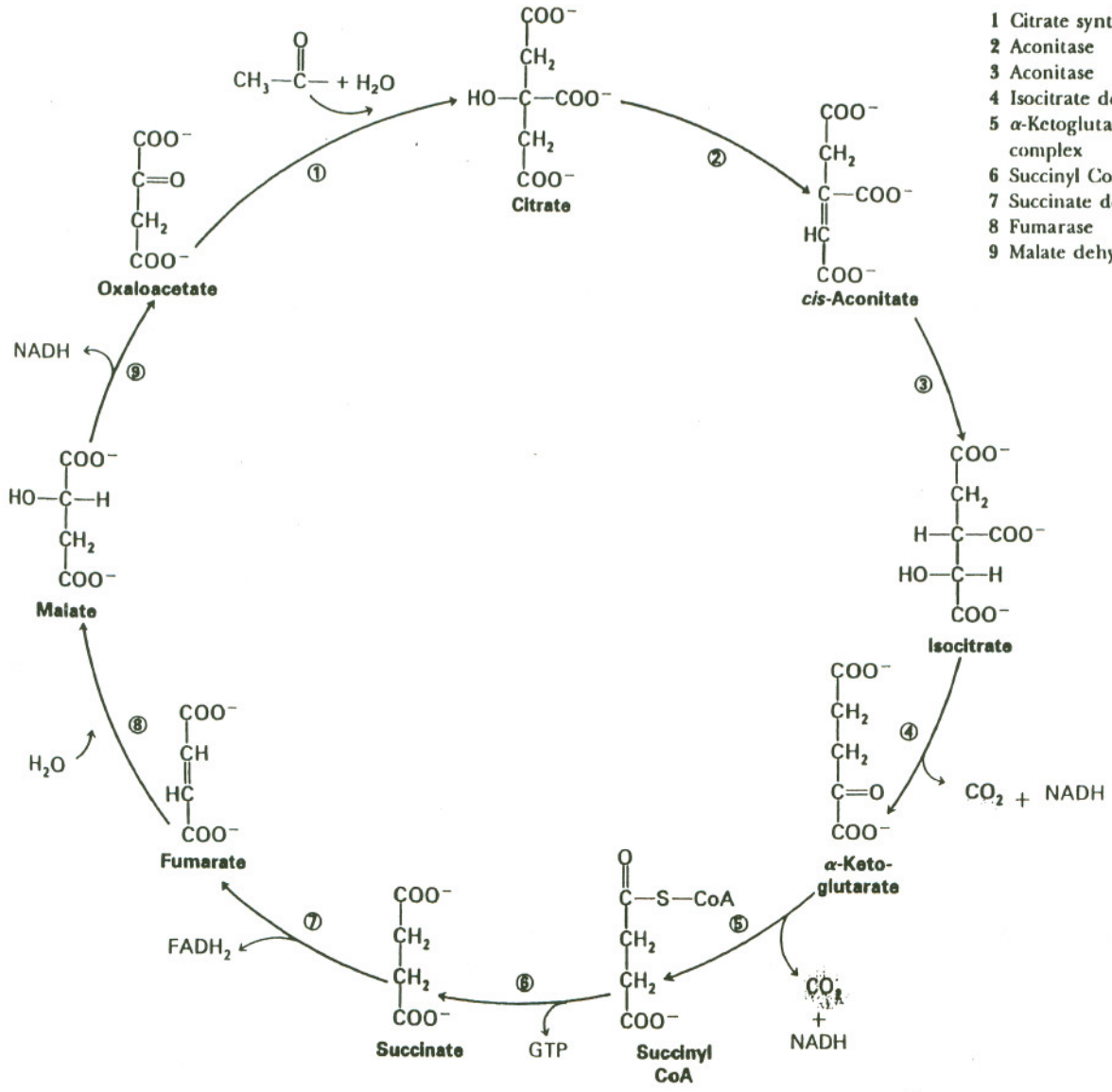


TABLE 12.1 Some naturally occurring fatty acids in animals

Number of carbons	Number of double bonds	Common name	Systematic name	Formula
12	0	Laurate	*n*-Dodecanoate	$CH_3(CH_2)_{10}COO^-$
14	0	Myristate	*n*-Tetradecanoate	$CH_3(CH_2)_{12}COO^-$
16	0	Palmitate	*n*-Hexadecanoate	$CH_3(CH_2)_{14}COO^-$
18	0	Stearate	*n*-Octadecanoate	$CH_3(CH_2)_{16}COO^-$
20	0	Arachidate	*n*-Eicosanoate	$CH_3(CH_2)_{18}COO^-$

following enzymes, as numbered in the diagram:

1 Citrate synthase
2 Aconitase
3 Aconitase
4 Isocitrate dehydrogenase
5 α-Ketoglutarate dehydrogenase complex
6 Succinyl CoA synthetase
7 Succinate dehydrogenase
8 Fumarase
9 Malate dehydrogenase



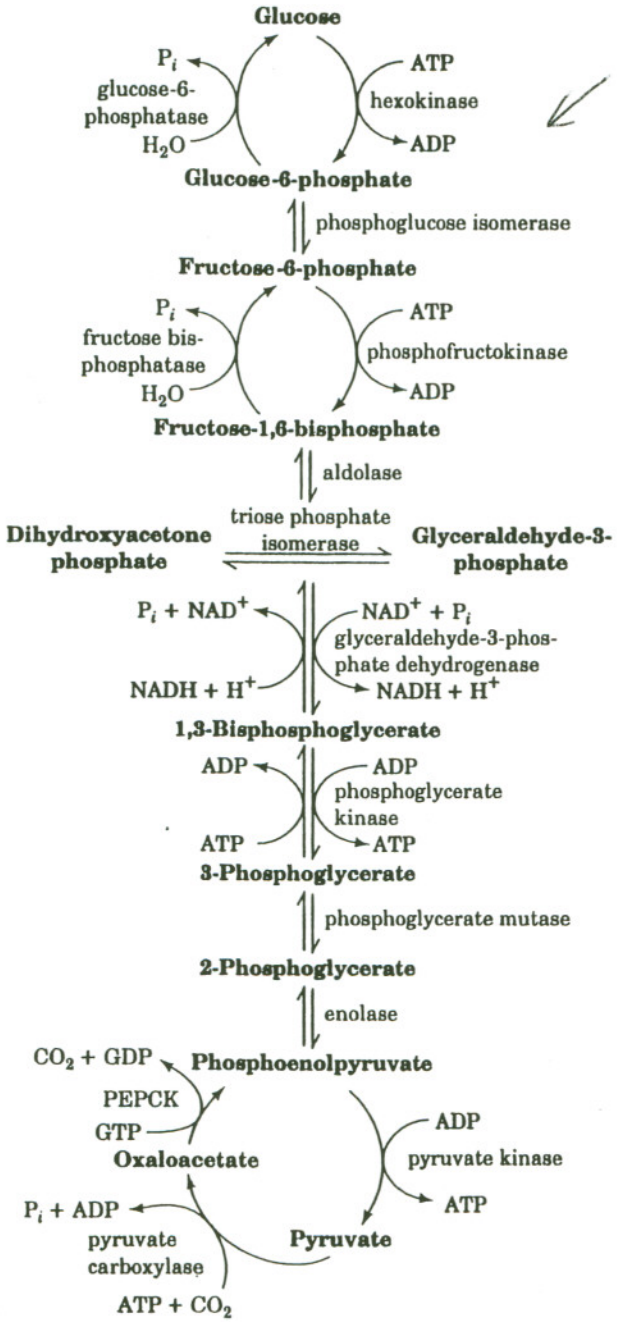
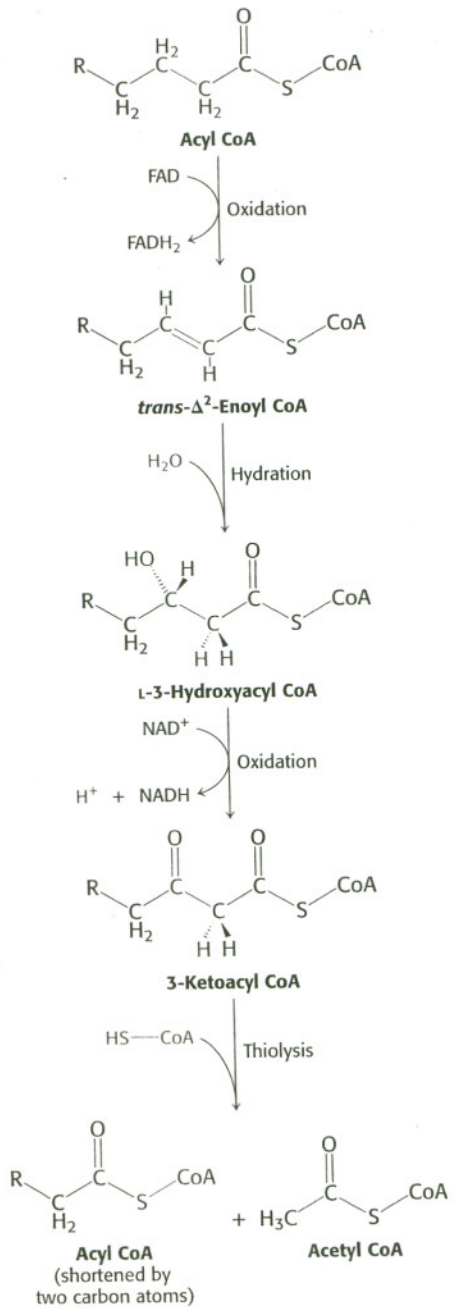
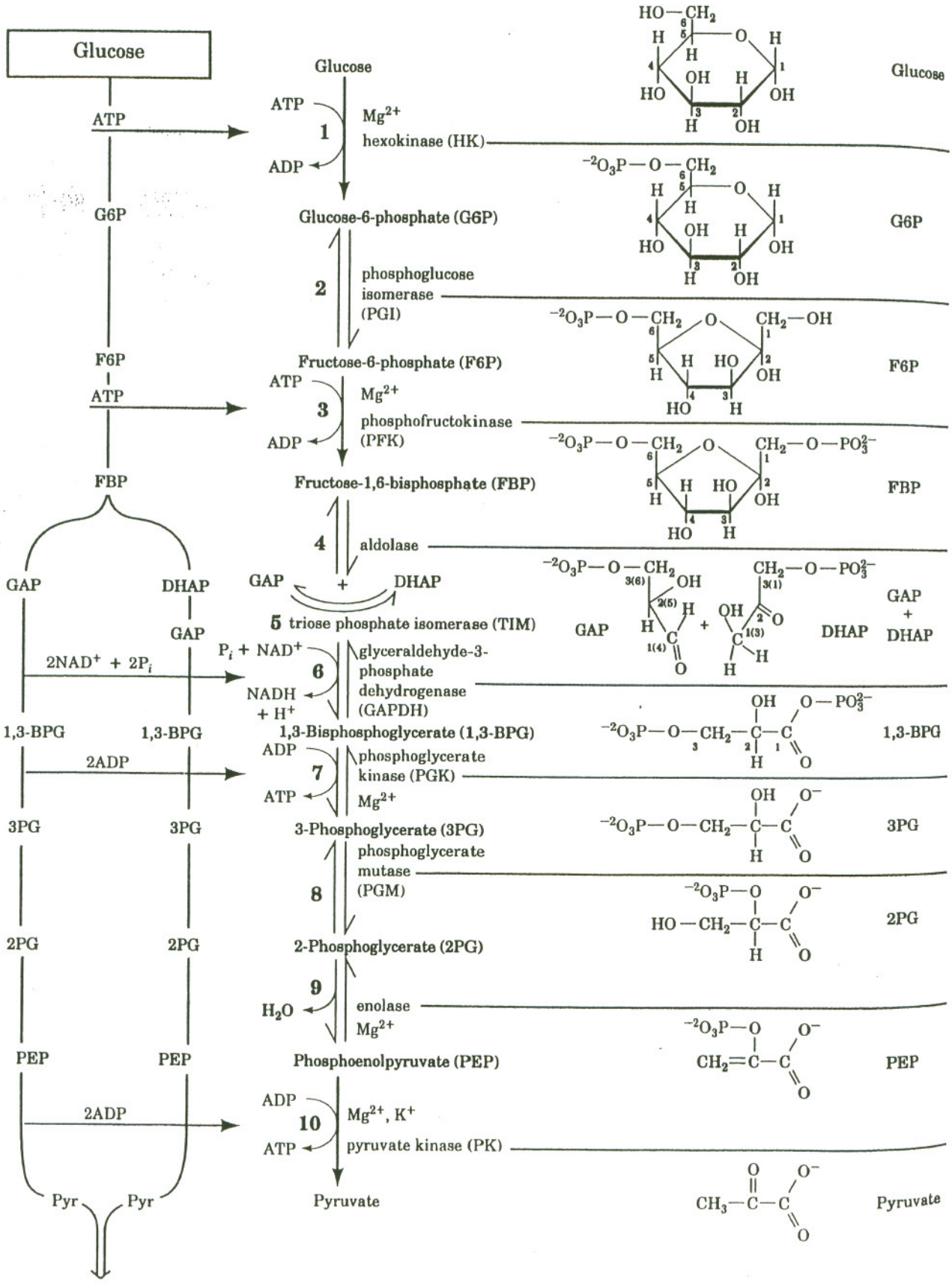


Figure 15-22. Key to Metabolism. **Comparison of the pathways of gluconeogenesis and glycolysis.** The red arrows represent the steps that are catalyzed by different enzymes in gluconeogenesis. The other seven reaction steps of gluconeogenesis are catalyzed by glycolytic enzymes that function near equilibrium.



Fatty Acid breakdown


Glucose
ATP
G6P
F6P
ATP
FBP
GAP
DHAP
GAP
2NAD+ + 2Pi
1,3-BPG
1,3-BPG
2ADP
3PG
3PG
2PG
2PG
PEP
PEP
2ADP
Pyr
Pyr
Glucose
ATP
1
Mg2+
hexokinase (HK)
ADP
Glucose-6-phosphate (G6P)
2
phosphoglucose isomerase (PGI)
Fructose-6-phosphate (F6P)
ATP
3
Mg2+
phosphofructokinase (PFK)
ADP
Fructose-1,6-bisphosphate (FBP)
4
aldolase
GAP
+
DHAP
5 triose phosphate isomerase (TIM)
Pi + NAD+
6
glyceraldehyde-3-phosphate dehydrogenase (GAPDH)
NADH + H+
1,3-Bisphosphoglycerate (1,3-BPG)
ADP
7
phosphoglycerate kinase (PGK)
ATP
Mg2+
3-Phosphoglycerate (3PG)
8
phosphoglycerate mutase (PGM)
2-Phosphoglycerate (2PG)
9
H2O
enolase
Mg2+
Phosphoenolpyruvate (PEP)
ADP
10
Mg2+, K+
pyruvate kinase (PK)
ATP
Pyruvate
Glucose
G6P
F6P
FBP
GAP
DHAP
GAP + DHAP
1,3-BPG
3PG
2PG
PEP
Pyruvate